Tutorial: Multi-Dataflow Composer Tool

CGR Coprocessor Generation for Kria KV260 Using MDC and PICEA–MLIR

1 Overview of the Design Flow

This tutorial describes how to generate and deploy an AXI-based processor–coprocessor system on the AMD/Xilinx Kria KV260 using MDC (http://sites.unica.it/rpct/) and, optionally, the PICEA–MLIR compilation flow (https://gitlab.abinsula.com/myrtus/dpe/picea). The associated deployment resources are available at https://github.com/mcheshfar/AXI-based-driver-applications.

The tutorial covers three alternative design flows:

1. **MDC-only flow:** provide the dataflow networks in XDF format, their CAL actor interfaces, and the corresponding HDL component library. MDC merges the networks and generates the coprocessor and platform-integration files.
2. **PICEA export with MDC merging:** start from an MLIR description and use PICEA to generate the XDF networks and CAL actor interfaces. The networks remain separate at this stage; MDC subsequently performs merging and generates the coprocessor and platform-integration files.
3. **PICEA export with MLIR-level merging:** start from an MLIR description and enable region merging in PICEA before exporting the resulting network to XDF/CAL. MDC then uses this already merged network for hardware composition and platform integration.

Two approaches to preparing the actor HDL library are described: the existing CAPH-based approach and the PICEA EmitHLS-based approach. In the latter, PICEA generates HLS-compatible C++ implementations and an actor-library Tcl script; executing this script with Vitis HLS synthesizes the actor HDL components and collects them with the required FIFO support files. The XDF/CAL descriptions specify the network structure and actor interfaces, while the HDL library supplies the corresponding hardware implementations.

All three flows converge on MDC for coprocessor generation and Kria integration. The generated Vivado scripts are used to build the FPGA bitstream, while the generated Linux package provides device-tree and driver sources together with userspace utilities for deployment and testing.

The running example is a multi-dataflow accelerator supporting two edge-detection algorithms: Sobel and Roberts. Each algorithm applies a pair of convolution kernels to a grayscale image (Figure 1) to identify intensity variations associated with image edges. The two algorithms serve as selectable configurations of the merged accelerator.

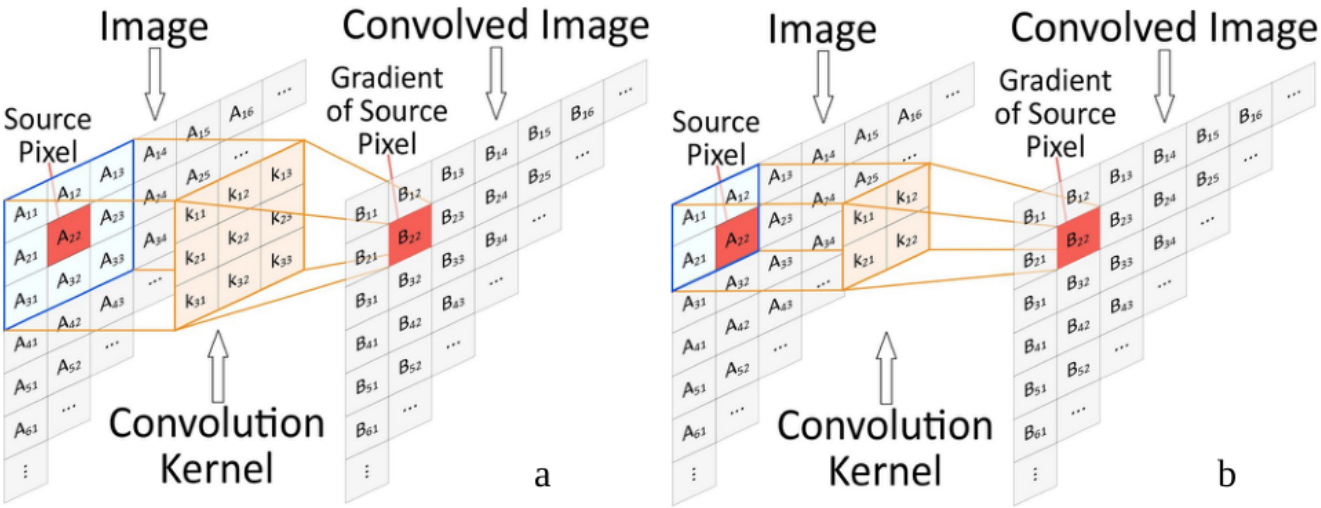
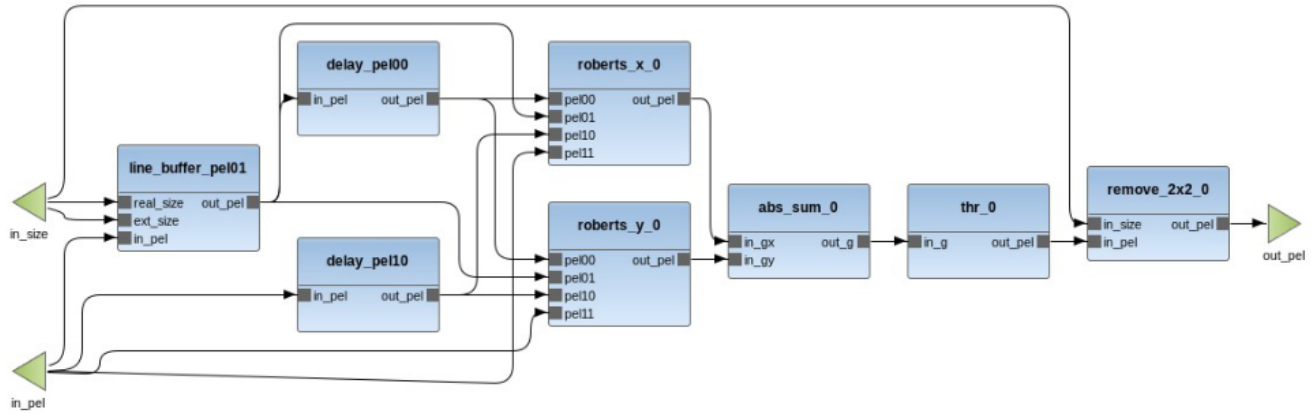
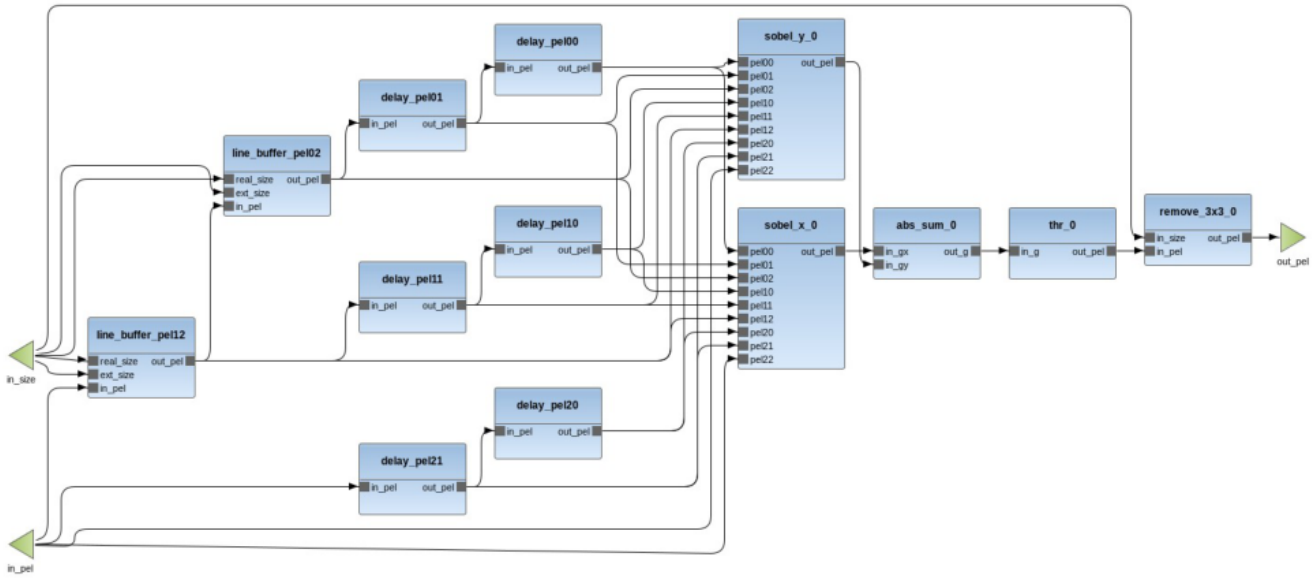


Figure 1: Computation of the convolution of kernels (x and y) of the Sobel (a) and Roberts (b) operators with an input image (A).

Figure 2 illustrates the single networks for Sobel (a) and Roberts (b) algorithms.



(a)



(b)

Figure 2: Dataflow description of the single networks for the Sobel (a) and Roberts (b) operators.

2 Preparing Networks and HDL Component Libraries

The individual dataflow networks and their HDL component libraries can be prepared using either CAPH or PICEA–MLIR.

CAPH-based preparation. CAPH (http://caph.univ-bpclermont.fr) is a framework for specifying, simulating, and implementing stream-processing applications based on a dynamic dataflow model of computation. In this approach, the network descriptions and corresponding HDL components are prepared using the CAPH-based flow and supplied to MDC for merging and coprocessor generation.

PICEA–MLIR-based preparation. Alternatively, PICEA (https://gitlab.abinsula.com/myrtus/dpe/picea) generates XDF network descriptions and CAL actor interfaces from an MLIR input. Region merging can either be performed within PICEA before export or deferred to MDC. For hardware-library generation, the PICEA EmitHLS flow produces HLS-compatible C++ actor implementations and a Tcl synthesis script. Executing this script with Vitis HLS generates the actor HDL library, including the required FIFO support files, for use by MDC. RESIN provides a common command-line entry point for network export and actor-synthesis input generation.

Necessary tools and resources:

- **MDC:** required for hardware composition, coprocessor generation, and Kria integration (use this version). When a prebuilt executable is not available, follow the MDC installation guide (docs.google.com).
- **Java Runtime Environment:** required to run MDC. Use a version compatible with the selected MDC/Eclipse distribution (v8 or higher).
- **AMD Vivado:** required to generate the FPGA bitstream from the MDC-generated hardware and Tcl scripts. This tutorial has been tested with Vivado 2024.2.
- **Kria KV260 board:** required for on-board execution, with a compatible Linux installation and the development tools and kernel headers needed to build the generated deployment software(here).
- **Example files:** the Sobel and Roberts network descriptions, actor HDL components, the corresponding MDC protocol description, and test inputs with reference outputs. For the PICEA-based flows, the MLIR source descriptions are also required (here).
- **CAPH, when using CAPH-based preparation:** required to regenerate the CAPH examples and their HDL components. It is not required when using the supplied network descriptions and HDL library.
- **PICEA–MLIR and RESIN, when using MLIR-based preparation:** built with the MDC export and EmitHLS functionality enabled (here).
- **AMD Vitis HLS, when using EmitHLS-based library generation:** required to synthesize the generated C++ actor implementations into HDL using the generated Tcl script (here).

Before running MDC, two complementary inputs must be available: the dataflow network descriptions and the HDL implementations of their actors. The XDF files describe actor instances and connections, while the CAL files define the corresponding actor interfaces. The HDL component library provides the hardware implementations used during MDC composition.

The following subsections describe three alternative preparation flows. Choose one of these flows according to the source representation and the stage at which network merging should be performed. For the PICEA-based flows, the final subsection explains how to synthesize the actor HDL library.

2.1 Flow 1: CAPH-Based Preparation for MDC Merging

This flow starts from the CAPH-based example package containing the individual Roberts and Sobel networks and their corresponding HDL components. PICEA and RESIN are not required for this path.

Prepare the following resources:

- the separate XDF network descriptions for Roberts and Sobel;
- the CAL actor interfaces referenced by these networks;
- the corresponding actor HDL implementations and their required dependencies;
- the MDC protocol description matching the interfaces of these HDL components.

When using the supplied example package, these resources are already available. CAPH is required only when regenerating or modifying the CAPH-based implementations.

Keep the two networks separate during this preparation stage as shown in Figure 2. Both networks will subsequently be supplied to MDC, which identifies compatible actors, performs network merging, and inserts the switching structures required to select between the Roberts and Sobel configurations.

The HDL library and protocol description must correspond to the same actor-interface convention. In particular, the data, handshake, clock, and reset signals described by the protocol must match the supplied hardware components.

2.2 Flow 2: PICEA–MLIR Export with Merging in MDC

This flow starts from an MLIR file containing the individual dataflow regions. PICEA prepares these regions and exports their network descriptions without merging them. MDC performs the merging step later.

From the PICEA repository root, run:

```
resin robertsobel.mlir --mdc --mdc-actors --hls-target-device=xck26-sfvc784-2LV-c -o build
```

The `--mdc` option enables MDC preparation and XDF/CAL export. The `--mdc-actors` option additionally generates the actor-synthesis inputs through EmitHLS. The `--hls-target-device` option specifies the FPGA device used for actor synthesis.

Do not enable `--merging` in this flow. The exported networks must remain separate so that MDC can perform the merging step.

The generated output contains Figure 3 files and directories:

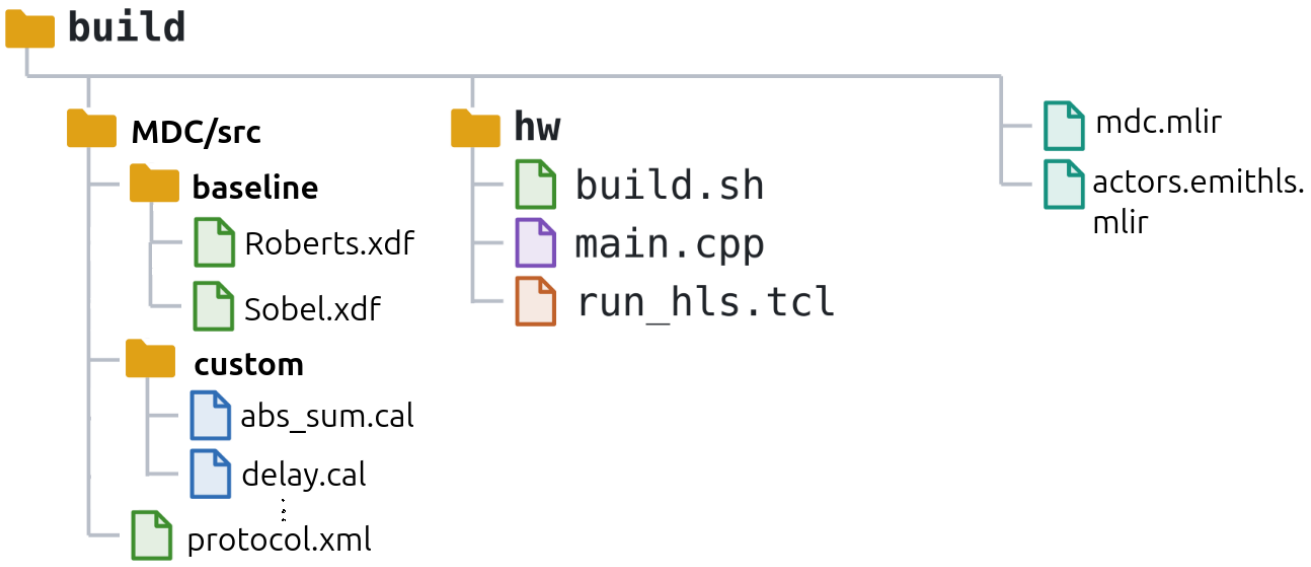


Figure 3: The generated output for resin without merging in PICEA

The file `mdc.mlir` contains the prepared DFG representation. The `MDC/src/baseline` directory contains the exported networks and their diagram files, while `MDC/src/custom` contains the CAL actor interfaces. Network filenames depend on the names defined in the input MLIR.

The file `actors.emithls.mlir` contains the representation used for actor code generation. The `hw/main.cpp` and `hw/run_hls.tcl` files are the inputs to the actor-library synthesis procedure described in Section 2.4.

After synthesizing the HDL library, supply the separate exported networks, the HDL components, and their matching protocol description to MDC.

2.3 Flow 3: PICEA–MLIR Region Merging and Export

This flow also starts from an MLIR description, but performs region merging within PICEA before XDF/CAL export. MDC then receives the already merged network and performs the remaining hardware-composition and platform-integration steps.

Enable MLIR-level merging by adding `--merging`:

```
resin robertsobel.mlir --mdc --mdc-actors --merging --hls-target-device=xck26-sfvc784-2LV-c \
-o build
```

The output follows the same directory organization as Flow 2 as shown in Figure 4. However, the prepared MLIR and exported network now represent the result of PICEA region merging, including the switching structures required by the merged graph. **The already merged network is the input to MDC for this flow.** Do not additionally provide the original separate networks as merging inputs, because this would mix the two alternative preparation paths.

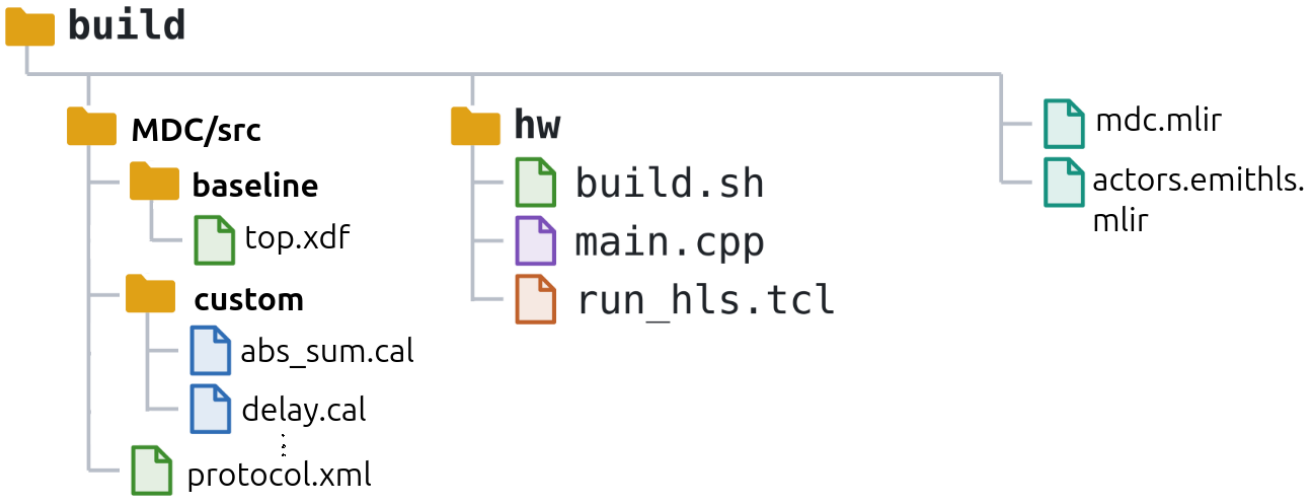


Figure 4: The generated output for resin without merging in PICEA

Generate the actor HDL library from the synthesis files produced by this same RESIN invocation. Keeping the exported network and actor library from the same prepared input avoids inconsistencies in actor names, interfaces, and automatically generated adapters.

The choice between Flow 2 and Flow 3 determines the location of the merging step. It does not imply that the two merging implementations necessarily produce identical graph structures or hardware-resource usage.

2.4 Generating Actor HDL Libraries with EmitHLS and Vitis HLS

For both PICEA-based flows, RESIN generates the source files required for actor synthesis. **RESIN does not execute Vitis HLS in the commands shown above.** HDL generation is performed by running the generated Tcl script explicitly.

For Flow 2 or Flow 3, execute:

```
cd build/hw
vitis-run --mode hls --tcl run_hls.tcl
```

Each command assumes that the Vitis HLS environment has been configured and that `vitis-run` is available in the shell.

The generated Tcl script creates a separate synthesis project for each selected actor and collects the resulting RTL in `HDL_components`. With the FIFO-support export update, this directory also contains the shared FIFO wrapper and its RTL dependency:

```
hw/
    main.cpp
    run_hls.tcl
    HDL_components/
        mdc_hls_fifo_native.v
        top_fifo_w32_d4_S.v
        <actor_1>/
        <actor_2>/
        ...
```

Actor-directory names depend on the input application. When preparation inserts additional actors, such as boundary packing and unpacking adapters, their implementations must also be included in the synthesized library.

The wrapper `mdc_hls_fifo_native.v` instantiates the FIFO defined in `top_fifo_w32_d4_S.v`. Both files must therefore remain available to the subsequent hardware-composition flow.

Regenerating the synthesis inputs manually. RESIN normally coordinates C++ and Tcl generation. For debugging or selective regeneration, the corresponding translation commands can also be executed directly. For example, from the PICEA repository root (remove –merging for Flow 2):

```
picea-opt robertssobel.mlir   --merging  --prepare-for-mdc    -o robertssobel.mdc.mlir

picea-opt robertssobel.mdc.mlir   --convert-to-emithls --for-mdc   -o actors.emithls.mlir

picea-translate robertssobel.mdc.mlir  --dfg-to-mdc --mdc-output-dir=build

picea-translate actors.emithls.mlir --emithls-to-cpp -o build/main.cpp

picea-translate actors.emithls.mlir  --emithls-to-hls-tcl --hls-actor-library \
--hls-source-file=main.cpp  --hls-target-device=xck26-sfvc784-2LV-c -o .build/run_hls.tcl
```

Both translations must use the same EmitHLS input. The actor-library exporter uses the generated actor inventory to select synthesis targets. An explicit `--hls-actors` argument overrides that inventory; omit it when the intention is to synthesize all recorded actors, including automatically inserted adapters.

Inputs to the next stage. After synthesis completes successfully, the PICEA-based preparation flow provides:

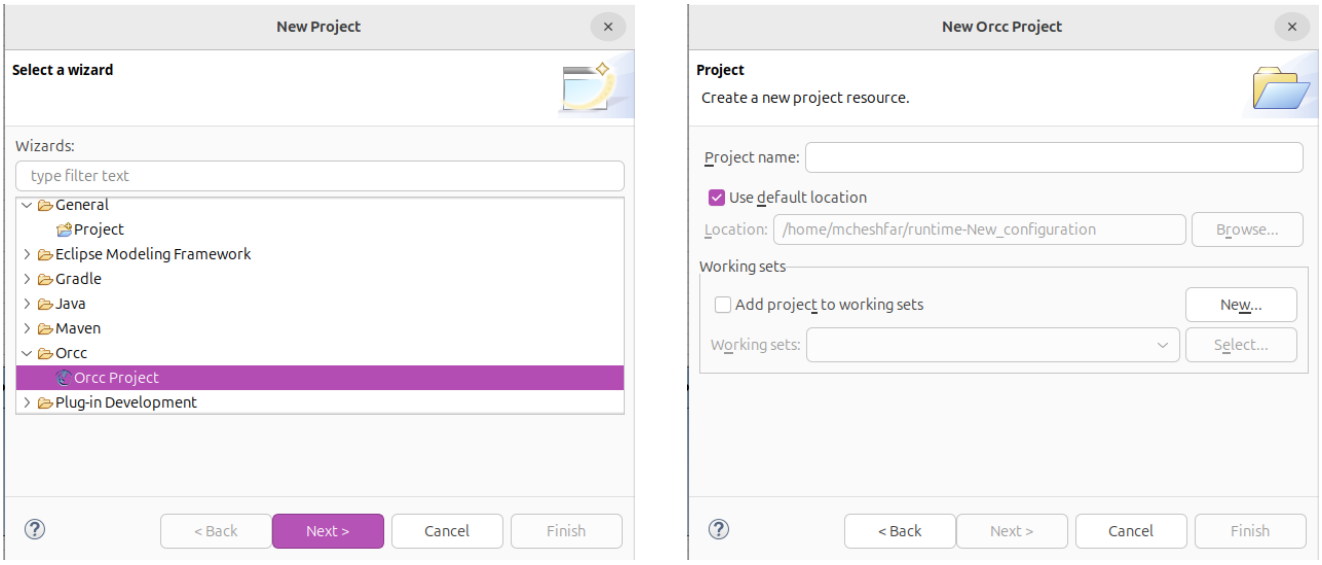
- the exported XDF networks and CAL actor interfaces;
- the synthesized actor HDL library and its dependencies;
- a matching FIFO protocol description, supplied separately for MDC integration.

These resources are used in the next section to configure MDC and generate the Kria coprocessor.

3 Configuring MDC for Coprocessor Generation

After preparing the networks and HDL component library, configure MDC to compose the accelerator and generate the Kria integration files. The three preparation flows converge at this stage. Their principal difference is whether MDC receives separate networks or a network already merged by PICEA.

Use a dedicated output directory for each experiment. Keeping the outputs of different flows separate simplifies validation and prevents files from previous runs from being mistaken for newly generated results.



(a) (b)

Figure 5: Create project steps.

3.1 Importing the XDF Networks and CAL Actor Interfaces

Run MDC tool (in Windows launch MDC executable, in Linux run it by eslipse) and import the prepared network project into the workspace. For the PICEA-based flows, use the generated `MDC` directory, preserving the source organization:

```
MDC/
    src/
        baseline/
            *.xdf
            *.xdfdiag
        custom/
            *.cal
```

The `baseline` directory contains the network descriptions, while `custom` contains their CAL actor interfaces. Preserve this organization so that actor references in the XDF files resolve correctly.

If the generated directory does not contain Eclipse project metadata, create an appropriate Orcc project in the workspace and copy the generated source tree into its source directory see figure 5 .

Refresh the project after copying or regenerating files. Open the XDF networks and check that all actor instances and connections are resolved before launching MDC. The `.xdfdiag` files support graphical visualization; the `.xdf` files are the network inputs to compilation.

Create or edit the MDC launch configuration and select the imported project. Add the required XDF files to the input-network list. The configured number of input networks must equal the number of XDF files selected.

For the Roberts–Sobel example, select both individual networks when merging in MDC like Figure 2. When using PICEA region merging, select only the exported merged network.

3.2 Configuring Network Merging or Using a Premerged Network

For Flow 1 and Flow 2, provide the separate Roberts and Sobel networks and select the MDC merging algorithm. When using the Moreano-based flow described in this tutorial, select the corresponding Moreano option.

MDC identifies compatible actor instances and connections, constructs the merged network, and generates the switching structures needed to activate each configuration. The resulting accelerator contains shared hardware where compatible operations can be reused.

For Flow 3, provide the single network already merged by PICEA. The input-network count is therefore one, even when that network represents multiple functional configurations.

The number of input XDF files and the number of runtime configurations are different concepts. A single premerged XDF can represent both Roberts and Sobel.

This path requires the MDC version supporting the PICEA-exported switching and configuration metadata. The existing switching structure and its mode information must be preserved during hardware generation. Selecting one XDF file alone does not establish that its internal configurations have been imported correctly.

Create a new run configuration: Run > Run configurations… , right click on Orcc compilation then select New. Now, chose a name for the configuration, and select Tutorial_EdgeDetection project your PICEA project, MDC backend and the output folder path, Figure 6.

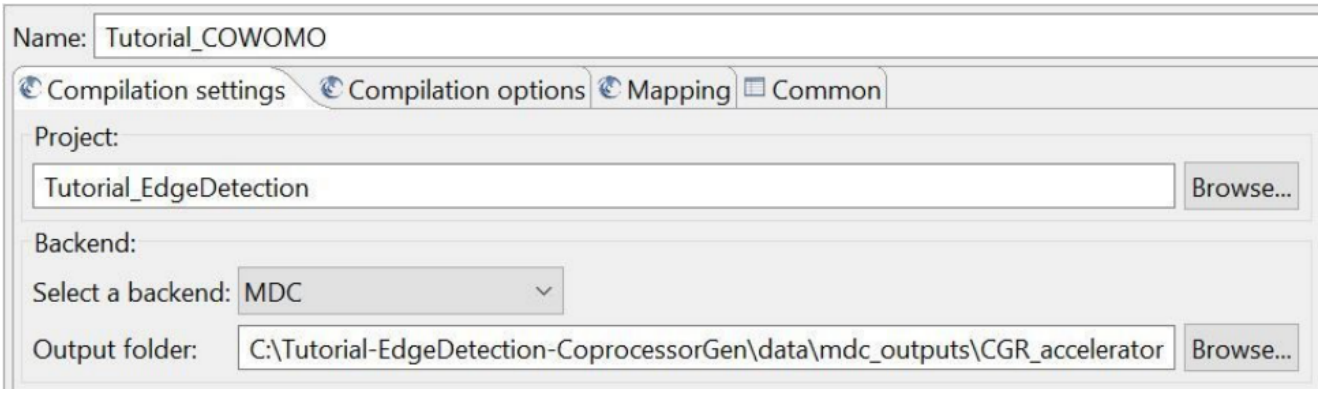


Figure 6: run configuration for select project

For Flow 1 and Flow 2, select the two input dataflow networks: edgeDetection.roberts and edgeDetection.sobel and choose the merging algorithm (EMPIRIC or MOREANO, ref). In this tutorial we adopt the Moreano one ,Figure 7. For Flow 3, enable Accept Merged Network and set 1 for number of networks.

After generation, inspect `configNetID.txt` and confirm that the expected configurations are present. Use the



Figure 7: merging options

generated mapping when selecting modes in the test application; do not assume that configuration IDs are identical across independently generated builds.

For merging networks Tick “Generate RVC-CAL multi-dataflow”. And Run.

This step merges the two input dataflow networks, through the selected datapath merging algorithm. Click on Run.

Refresh the project folder to visualize the output folder with the generated multi-dataflow. In the generated networks you can notice as actors are shared, and functionalities of the two input networks are guaranteed by the insertion of the switching boxes as shown in Figure 8.

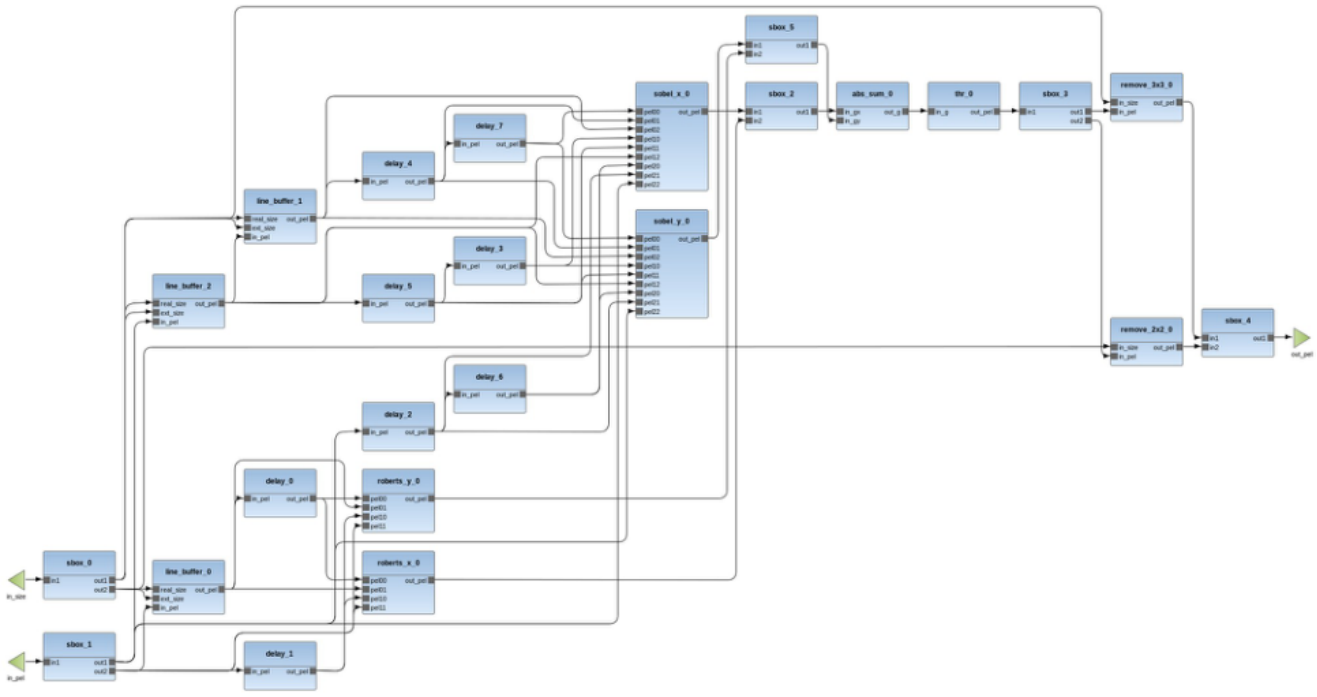


Figure 8: merged network

3.3 Selecting the HDL Library and Protocol Description

Open again the configuration window: Run > Run configurations... Choose previous configuration, unselect “Generate RVC-CAL multi-dataflow” and tick “Generate HDL multi-dataflow” as Figure 9. Enable HDL generation and specify the HDL component library and its matching protocol description.

For the CAPH-based flow, select the supplied CAPH HDL library and the corresponding CAPH protocol file. Include the supporting VHDL packages and FIFO modules required by the actor implementations.

For the EmitHLS-based flow, select the `hw/HDL_components` directory produced by the Vitis HLS script. For

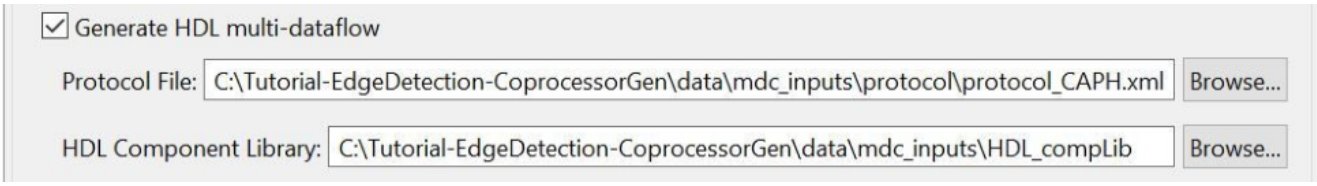


Figure 9: The HDL component library options

example:

```
build/hw/HDL_components
```

Select `protocol_emitHLS.xml` as the matching protocol description. Use its location in the prepared example package or generated MDC project, as applicable to the RESIN version being used.

The protocol description tells MDC how to connect the actor data ports, handshake signals, and system signals. It must match the interfaces implemented by the selected HDL library.

The CAPH and EmitHLS libraries require their corresponding protocol descriptions. A network with correct actor names can still produce incorrect hardware connections if the selected protocol does not match the actor interfaces.

3.4 Configuring the Kria KV260 Target

To generate the Ip, as shown in Figure 10 enable **System Generation** and **Generate Accelerator IP** for coprocessor generation. For the streaming coprocessor flow used in this tutorial, select the `STREAM` interface type and `ARM` for Host Processor. Enable `Adop DMA`.



Figure 10: Coprocessor Generation options

Enter the following target identifiers:

```
Board part: xilinx.com:kv260_som:part0:1.4
Part name:  xck26-sfvc784-2LV-c
```

The board-part identifier must be available in the Vivado installation used for the build. The FPGA part must also agree with the device selected during Vitis HLS actor synthesis.

The board and part fields specify the hardware target; they do not replace the generation-enable options. Apply the launch configuration before running MDC.

Choose a fresh output directory, for example `MDCoutput`, and launch generation. Review the console for unresolved actors, missing HDL dependencies, protocol errors, or generation failures.

Successful MDC execution produces the accelerator sources and platform-integration files. **The FPGA bitstream is generated subsequently by executing the Vivado build flow.**

4 Understanding the MDC-Generated Files

The generated output separates the accelerator implementation, Vivado integration, and Linux software, as shown in Figure 11. These groups serve different stages of the deployment process and should be retained together as one build package.

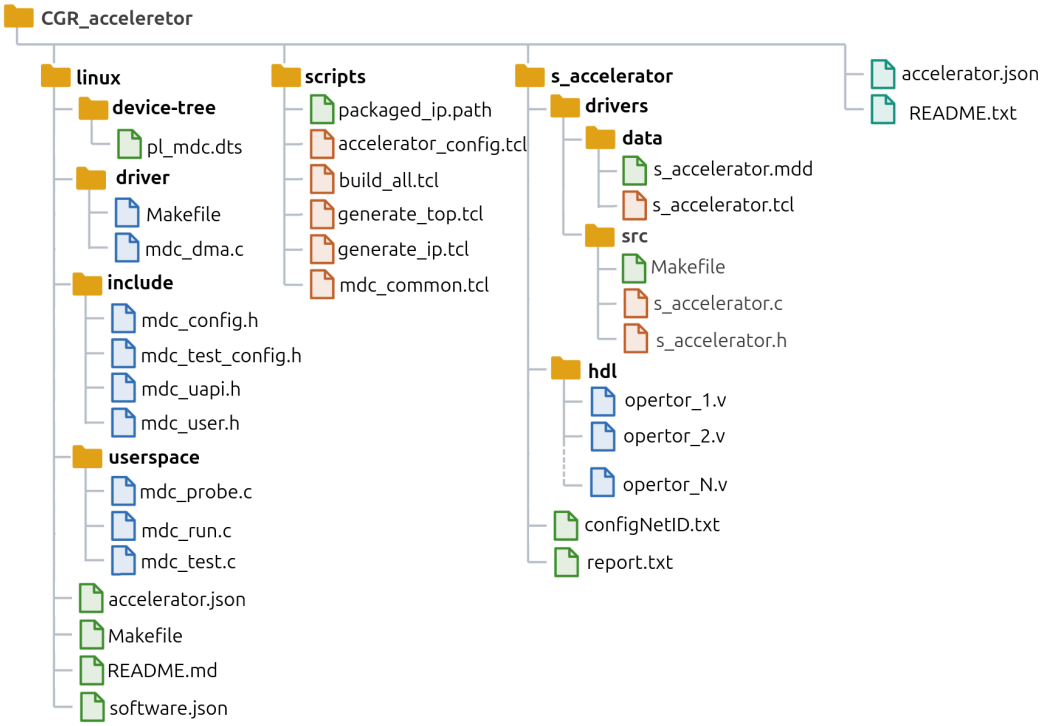


Figure 11: Coprocessor Generation options

The following subsections describe the principal outputs of the updated Kria flow. Some additional files and directory names depend on the enabled generation options and the MDC revision.

The accompanying `report.txt` records information about network composition and sharing. It is not a substitute for Vivado utilization and timing reports. The `configNetID.txt` file maps network configurations to the IDs used for mode selection.

4.1 Accelerator HDL and Packaged IP

For the streaming coprocessor, the `s_accelerator` directory contains the generated accelerator hardware. The composed dataflow network is represented by `multi_dataflow.v`, together with the actor implementations, switching logic, FIFO support, and interface modules required by the selected configuration.

The generated wrapper connects the dataflow network to the coprocessor interfaces. The actor-level FIFO protocol and the processor-facing AXI interfaces serve different roles: the former connects computational components inside the accelerator, while the latter supports system integration and processor communication.

The packaged accelerator IP is organized under `acc_ip`. Its principal contents include:

`hdl/` The RTL sources and dependencies included in the packaged accelerator.

`component.xml` The IP-XACT description used by Vivado to identify the IP, its interfaces, parameters, and source files.

`xgui/` Tcl definitions for the IP customization interface in Vivado.

`drivers/` Software-driver sources and metadata associated with the packaged IP, typically organized into `src` and `data` directories.

The packaged-IP driver directory is distinct from the Linux kernel-driver package. The Linux deployment described in this tutorial uses the separately generated `linux` directory.

Keep the packaged IP directory intact when moving the build to another machine. Its metadata references the accompanying HDL and support files.

4.2 Vivado Project and Build Scripts

The generated Tcl scripts automate the integration of the accelerator into the Kria hardware system. Their responsibilities include registering or packaging the accelerator IP, creating the Vivado project, constructing the block design, and configuring the required processing-system and AXI connections.

The platform-generation flow also establishes the clocks, resets, address assignments, and DMA connectivity used by the generated software configuration.

Use the generated top-level build script as the entry point. Preserve the relative locations of the scripts and packaged IP, because the scripts may resolve source paths relative to the generated output directory. you can run script on terminal in MDC output directory.

```
vivado -mode batch   -source scripts/build_all.tcl
```

Vivado execution creates additional build products, including the project files, block-design products, synthesis and implementation results, reports, and the FPGA bitstream. You can find `design_1_wrapper.bit` in build folder. Moreover,another option is to execute scripts by running Vivado as far too.

Launch Vivado.

Change directory to the root folder of the generated scripts, as: .../data/mdc_outputs/CGR_coprocessor

Now, in the main menu (only in Linux environment): Tools → Run tcl Script... and choose `.../CGR_coprocessor/scripts/build_all.tcl`. (Figure 12)

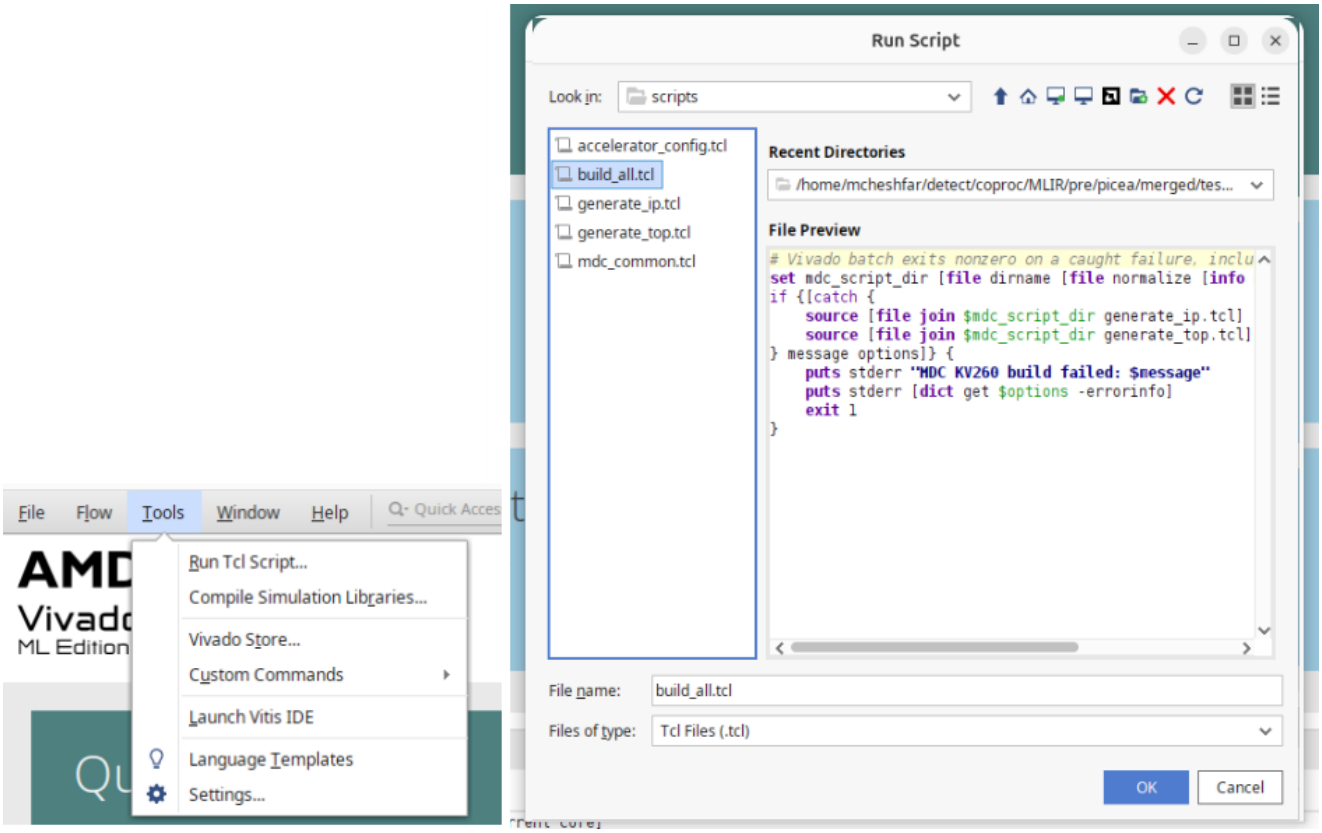


Figure 12: select tcl to run

It make bitstream file automatically. You can file `design_1_wrapper.bit` in build/top/run-**... .

Open the Vivado project: vivado build/top/run-**/mdckv260.xpr (Figure 13)

These Vivado build products are created after MDC generation. Their absence immediately after an MDC run does not indicate that coprocessor generation has failed.

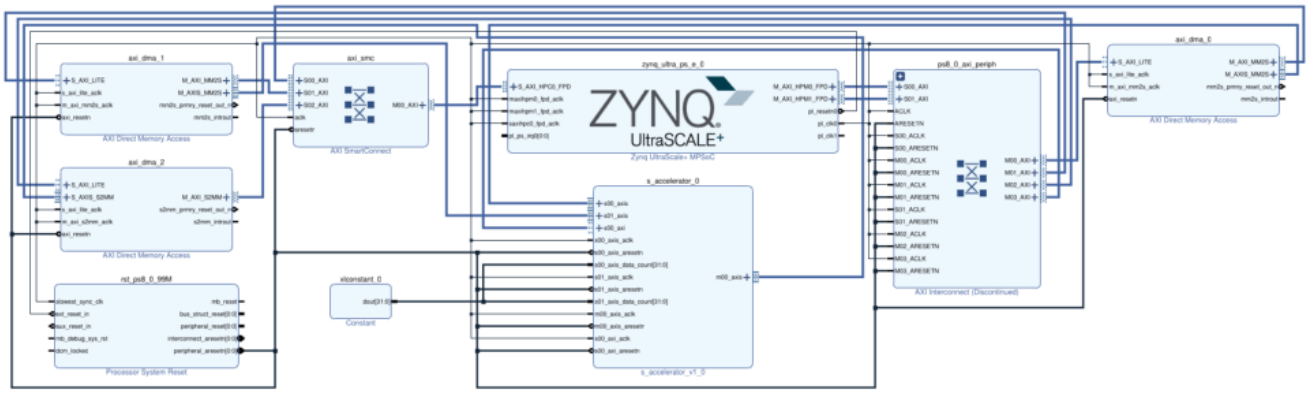


Figure 13: Block design for coprocessor

Inspect the Vivado timing and utilization reports before deployment. A generated bitstream establishes that implementation completed; numerical correctness must still be checked on the board using reference input and output data.

4.3 Linux Device Tree, Driver, and Userspace Utilities

The `linux` directory contains the software and hardware-description sources used to access the accelerator from Kria Linux. Its main components are the device-tree source, kernel driver, userspace utilities, and build rules.

Device-tree source. The generated `pl_mdc.dts` describes the programmable-logic resources required by the Linux deployment, including the relevant register regions and driver-binding information.

The source is compiled into a device-tree overlay for deployment. The overlay must correspond to the hardware implemented by the loaded bitstream.

Kernel driver. The `driver` directory contains `mdc_dma.c` and its generated configuration header, `mdc_config.h`. The driver provides the Linux interface used by the userspace utilities to control accelerator execution and DMA transfers.

Although the driver implementation is reusable, the configuration header contains application-dependent information, including channel organization, supported modes, buffer settings, and expected register addresses.

Consequently, **the same driver source does not imply that one compiled kernel module can be reused for every accelerator**. Build `mdc_dma.ko` using the configuration generated for the selected hardware and headers compatible with the running Kria kernel.

Userspace utilities. The `userspace` directory contains the generated utilities for inspecting and testing the accelerator. These include `mdc_probe` and `mdc_test`, together with the corresponding application configuration.

The generated `mdc_test_config.h` describes the available ports, token widths, modes, and active interfaces. The test application uses this information to interpret named inputs and outputs, select a configuration, submit transfers, and compare results against supplied reference files.

4.4 Network Configuration IDs and Generation Reports

MDC generates `configNetID.txt` to associate each functional configuration with its runtime identifier. For the merged Roberts–Sobel accelerator, these identifiers select the switching configuration required to execute either Roberts or Sobel on the shared hardware.

Inspect this file before running the accelerator and use the mapping produced by the current build. Configuration identifiers should not be assumed to remain unchanged when the input networks or their ordering change. When using the generated `mdc_test` utility, configurations can be selected by the mode names recorded in its generated configuration, for example through `--mode Sobel`.

For a network merged within PICEA, verify that the generated mapping retains the original functional configurations. Although MDC receives a single merged XDF file, the accelerator can still expose multiple selectable modes.

The generated `report.txt` summarizes network composition and actor sharing. It helps identify which actors are shared between configurations and which remain specific to an individual network.

The generation report describes the composed network; it does not establish hardware performance or functional correctness. Obtain FPGA resource usage and timing results from Vivado reports, and verify numerical correctness by comparing accelerator outputs with the corresponding reference data.

Retain `configNetID.txt` and `report.txt` with the generated hardware and Linux software so that the configuration mapping and composition results remain traceable to the deployed build.

5 OS-ORIENTED ACCELERATION MANAGEMENT ON KRIA KV260

1. [on your laptop/PC] SD Flashing —> (KV260 Ubuntu 22.04 image)



Figure 14: flash SD card with application on PC

2. Connect the board (via Ethernet or USB, or with keyboard/screen), insert the SD, and turn the board on

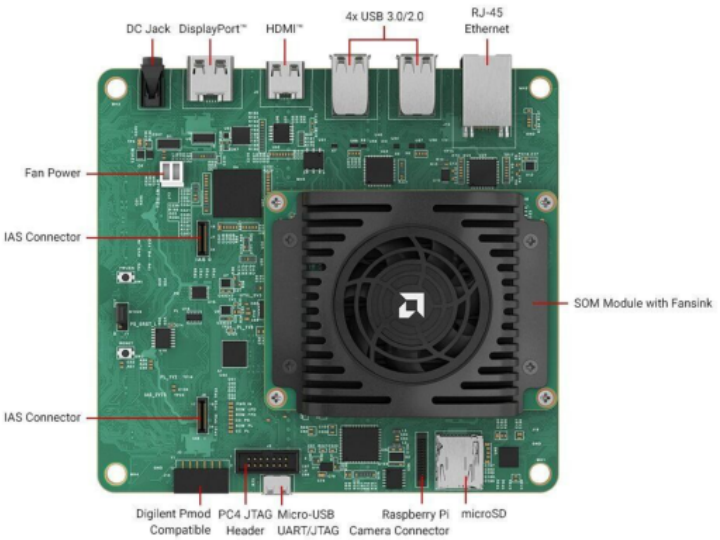


Figure 15: Kria KV260 board

3. (optional) [on your laptop] Control the board from via USB:

- Windows laptop: install Putty and open a serial connection (baudrate: 115200)
- Linux laptop: install picocom and run the following command (check if ttyUSB1 is related to the board): sudo picocom -b 115200 /dev/ttyUSB1

4. [on the board] Access the OS (user/pw: ubuntu) and change password (at least 8 characters)

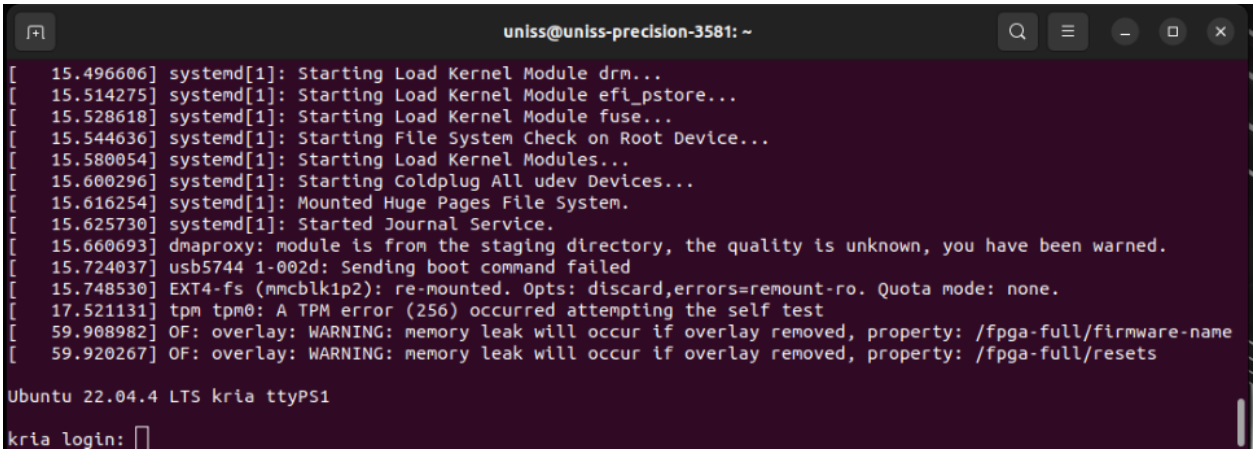


Figure 16: login in Kria Ubuntu

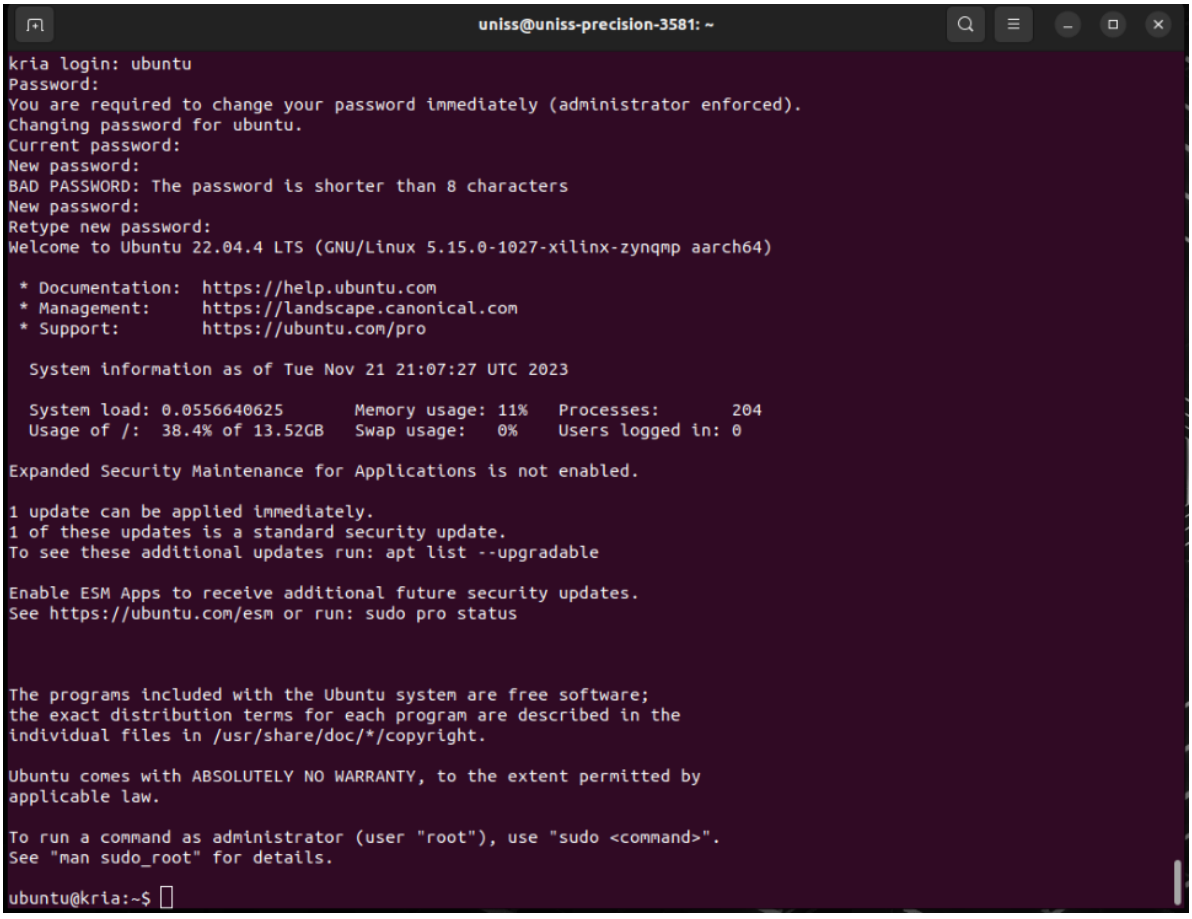


Figure 17: Kria Ubuntu is ready to use

5. (optional) Control the board from via Ethernet:

- [on the board] get the IP with the command (Figure 18): ifconfig
- [on your laptop] access through SSH (the IP must be the same you get with ifconfig)

```
ssh ubuntu@192.168.178.70
```

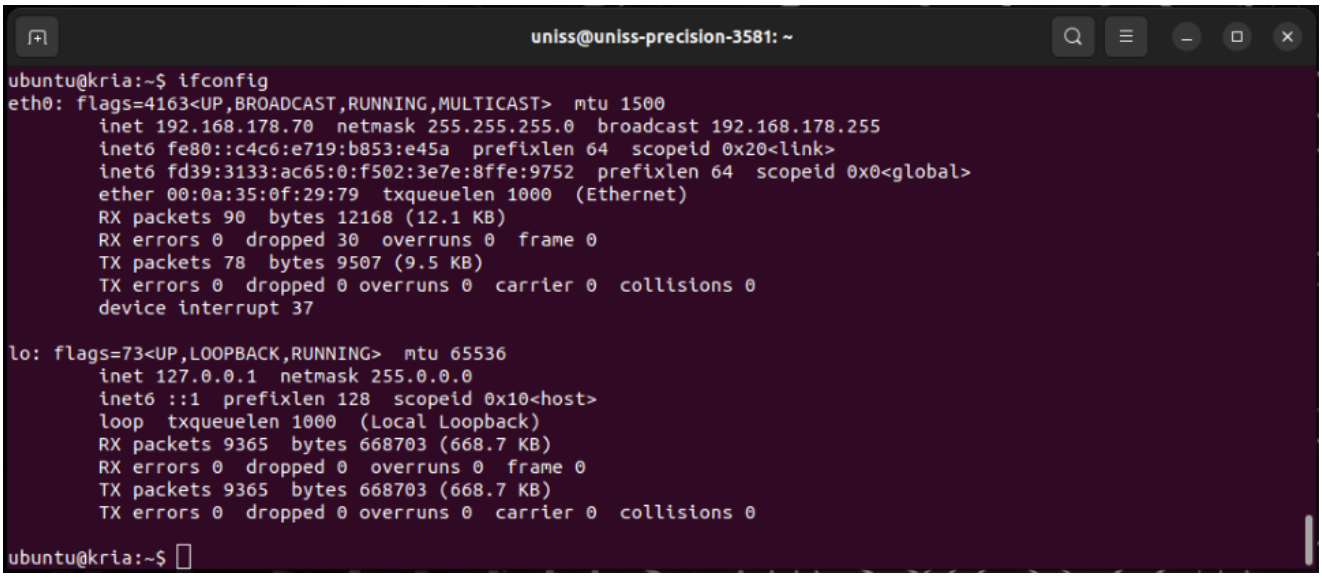


Figure 18: get the IP with the command

6 Deploying the Coprocessor on Kria Linux

Deployment requires the generated FPGA bitstream, device-tree overlay, Linux kernel driver, and userspace utilities. These files must correspond to the same accelerator build so that register addresses, channel directions, and configuration identifiers remain consistent.

Copy the generated Linux package and the corresponding bitstream to the Kria board. The following commands are executed on Kria unless otherwise stated. Replace the example deployment paths with the actual file locations.

6.1 Building the Device-Tree Overlay and Linux Software

Open a terminal on Kria and enter the generated `linux` directory:

```
cd /path/to/MDCoutput/linux
```

The board requires a C compiler, Make, the device-tree compiler, and kernel headers matching the running Linux kernel. Check the kernel version and its build directory:

```
uname -r
ls /lib/modules/$(uname -r)/build
```

Build the userspace utilities and device-tree overlay:

```
make userspace overlay
```

The overlay target compiles the generated device-tree source into a `.dtbo` file. The userspace target builds the accelerator inspection and test utilities, including `mdc_probe` and `mdc_test`.

Build the kernel module against the running kernel:

```
make module KDIR=/lib/modules/$(uname -r)/build
```

The resulting module is `driver/mdc_dma.ko`. It incorporates the application-specific settings from `mdc_config.h`.

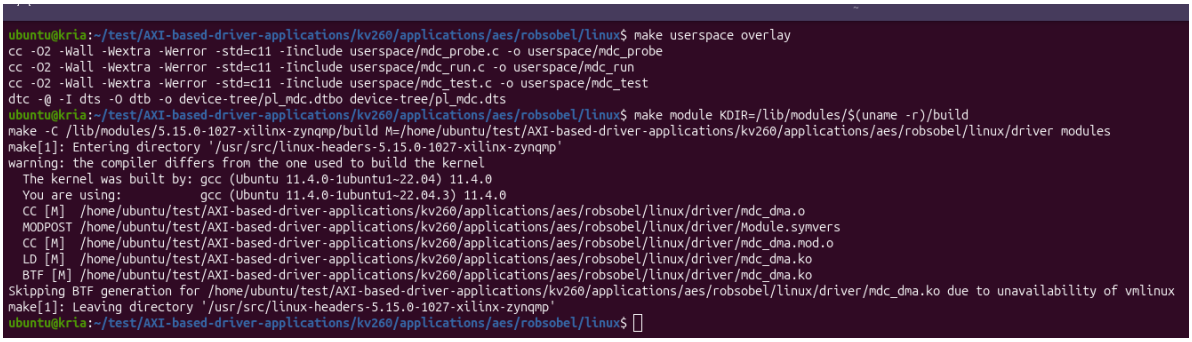


Figure 19: building usersapce and kernel example

Rebuild the driver and userspace utilities when the generated accelerator configuration changes. A module built for another kernel or another channel configuration must not be assumed compatible. If the kernel build directory is missing, install or prepare the matching kernel headers before continuing.

6.2 Loading the Bitstream, Overlay, and Driver

The following procedure uses `fpgautil` to load the programmable-logic configuration and apply the corresponding device-tree overlay through Linux FPGA Manager. It assumes that `fpgautil` and overlay support are available in the board's Linux image.

Prepare the board. Stop any application using the accelerator before replacing its hardware configuration. If a previous `mdc_dma` module is loaded, unload it after all clients have closed the device:

```
lsmod | grep mdc_dma
sudo rmmod mdc_dma
```

Run the removal command only when the module is loaded. Release any previously loaded FPGA application and its overlay using the mechanism that loaded them.Do not load a second configuration over an active application.

Load the hardware and overlay. From the generated `linux` directory, execute the following command after replacing both paths:

```
cd ..
sudo fpgautil -R
sudo fpgautil -b design_1_wrapper.bit
sudo fpgautil -o linux/device-tree/pl_mdc.dtbo
```

The binary bitstream and overlay must describe the same hardware system. The overlay must also be compatible with the base device tree of the running Linux image.

Check the command result and kernel messages before loading the driver:

```
sudo dmesg | tail -n 60
```

Do not continue if FPGA programming or overlay application reports an error.

Load the kernel driver. Insert the module built for this accelerator:

```
sudo insmod linux/driver/mdc_dma.ko
```

After successful device binding, verify that the accelerator device is available (Figure 20):

```
ls -l /dev/mdc_accel0
ls /sys/class/misc/mdc_accel0
sudo ./linux/userspace/mdc_probe
```



Figure 20: verification of the accelerator device available example

Loading the module alone does not establish successful device binding. The driver also requires the matching device-tree node and accessible hardware resources before it can expose `/dev/mdc_accel0`.

If the device node is missing, inspect the kernel log:

```
sudo dmesg | tail -n 100
```

Check for overlay failures, driver probe errors, resource conflicts, or discrepancies between the device-tree register addresses and the generated driver configuration. Creating the device node manually does not resolve these binding failures.

Once `mdc_probe` completes successfully, the system is ready for functional testing with `mdc_test`. The next section describes how to select Roberts or Sobel, transfer input data, and compare the accelerator outputs with reference results.

7 Testing and Validating the Accelerator

After loading the hardware, applying the device-tree overlay, and verifying driver access, use the generated `mdc_test` application to execute each configuration and compare its output with reference data.

The commands below reproduce the previously validated 16×16 edge-detection test. They are executed from the deployment directory containing the `linux` folder and the input and reference files.

7.1 Preparing Input Data and Reference Outputs

Prepare the input image, the image-size parameter, and the reference outputs for both algorithms. The example uses:

`in_size_file.mem` The size parameter supplied through input port `in0`.

`in_pel_file.mem` The image pixels supplied through input port `in1`.

`out0_Roberts_file.mem` The reference output for Roberts.

`out0_Sobel_file.mem` The reference output for Sobel.

Both configurations use the same input image and size parameter in this test. Their reference outputs are different you can find the examples here (link).

Use the memory-file format accepted by the generated test application. These files contain numerical tokens; an image file such as PNG or JPEG cannot be supplied directly without conversion.

The reference computation must use the same pixel ordering, arithmetic widths, signedness, border handling, and output ordering as the hardware. For this example, Roberts produces $15 \times 15 = 225$ output tokens and Sobel produces $14 \times 14 = 196$ output tokens, corresponding to valid processing without output padding.

Reference data must correspond to the selected configuration and input image. A successful comparison against an unrelated reference does not establish correctness of the intended computation.

7.2 Selecting and Testing the Roberts Configuration

Select the Roberts configuration using its generated mode name, associate the input files with their ports, and specify both the output destination and reference:

```
sudo ./linux/userspace/mdc_test  --mode Roberts  --input in0=in_size_file.mem \
  --input in1=in_pel_file.mem  --output out0=out0_Roberts_actual.mem \
  --expect out0=out0_Roberts_file.mem
```

Use the mode spelling recorded in the generated configuration. In the validated build, the Roberts mode is named `Roberts` and has identifier 1.

The successful test produced:

```
DMA transfers submitted for mode Roberts (id=1)
in0 completed: status=0x00001002 bytes=4
in1 completed: status=0x00001002 bytes=1024
out0 completed: status=0x00001002 bytes=900
out0: reference verified (225 tokens)
```

The output is saved to `out0_Roberts_actual.mem`. The reference-verification message confirms that the application compared the returned tokens with the supplied Roberts reference successfully.

7.3 Selecting and Testing the Sobel Configuration

Select Sobel using the same input files and its corresponding reference:

```
sudo ./linux/userspace/mdc_test --mode Sobel  --input in0=in_size_file.mem \
  --input in1=in_pel_file.mem  --output out0=out0_Sobel_actual.mem \
  --expect out0=out0_top2_file.mem
```

In the validated build, the Sobel mode is named `Sobel` and has identifier 2. Confirm the mapping in the current generated configuration instead of assuming that these identifiers apply to every build.

The successful test produced:

```
DMA transfers submitted for mode Sobel (id=2)
in0 completed: status=0x00001002 bytes=4
in1 completed: status=0x00001002 bytes=1024
out0 completed: status=0x00001002 bytes=784
out0: reference verified (196 tokens)
```

The output is saved to `out0_Sobel_actual.mem`. The smaller output count reflects the larger Sobel kernel and the border-handling convention used by this example.

7.4 Verifying Numerical Results

Validation requires successful data transfers and agreement with the numerical reference. **DMA completion alone does not demonstrate that the accelerator computed the correct result.**

For the validated 16×16 example, the expected transfer sizes are:

Transfer	Tokens	Bytes
Size parameter (`in0`)	1	4
Input image (`in1`)	256	1024
Roberts output (`out0`)	225	900
Sobel output (`out0`)	196	784

These byte counts correspond to four-byte transport words in this build. They should not be inferred solely from the internal pixel width or reused unchanged for other interface configurations.

Confirm that each execution reports the expected output-token count and a successful reference comparison. If a transfer fails or the numerical comparison reports a mismatch, check the selected mode, input-file ordering, reference file, transfer sizes, and consistency of the deployed hardware and software package.

Repeat each configuration with the same input, then alternate Roberts and Sobel without reloading the bitstream. Compare every execution against its reference to check repeated operation and configuration switching, including possible effects of retained actor or FIFO state.

Retain the input files, reference outputs, actual outputs, console logs, and build identifiers as the reproducible validation record.

Links

http://sites.unica.it/rpct/

https://gitlab.abinsula.com/myrtus/dpe/picea

https://github.com/mcheshfar/AXI-based-driver-applications

http://caph.univ-bpclermont.fr

https://github.com/mdc-suite/mdc/tree/mlir-frontend

https://docs.google.com/document/d/1G5cKw_gC97QreTQf8Wt6QaS4xfNMwW3k0dqt0z6wVLc/edit?tab=t.0#heading=h.s5si1wegx22v

http://www.oracle.com/technetwork/java/javase/downloads/jre8-downloads-2133155.html

https://www.amd.com/en/products/system-on-modules/kria/k26/kv260-vision-starter-kit.html

https://github.com/mcheshfar/AXI-based-driver-applications/tree/main/kv260/applications/edgedetection/MDC/MDC

https://gitlab.abinsula.com/myrtus/dpe/picea/-/tree/dec-mdc?ref_type=heads

https://www.amd.com/en/support/downloads/adaptive-socs-and-fpgas/development-tools/2026-1.html

https://www.researchgate.net/publication/278698706_CAPH_A_language_for_implementing_stream-processing_applications_on_FPGAs

https://drive.google.com/drive/folders/1au0F4E8QVDqZaL0TMWrrsKMHWlGEcQQM

https://drive.google.com/drive/folders/119dGOHhdyoGmOHCXryL6khIiJ3Ntfl9X

http://caph.univ-bpclermont.fr/

https://ieeexplore.ieee.org/document/1227149/

https://www.xilinx.com/support/documents/ip_documentation/axi_ref_guide/latest/ug761_axi_reference_guide.pdf

https://people.canonical.com/~platform/images/xilinx/kria-ubuntu-22.04/iot-limerick-kria-classic-desktop-2204-20240304-165.img.xz